Janghyun **Noh**

AI Researcher

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Education

Myongji University
Seoul, S.Korea

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Thesis: *Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data*
- Advisor: Prof. Minsik Oh, Artificial Intelligence and Data Analytics (AIDA) Lab
- GPA: 4.44 / 4.5
- Award: Research Honor Award from the President of Myongji University (at graduation)
- Teaching Assistant: Deep Learning, Machine Learning, Data Mining, Basic Programming (Fall 2024 – Spring 2026)

Myongji University
Seoul, S.Korea

B.S. in Data Technology, School of Convergence Software — *Mar. 2018 - Aug. 2024*

- Capstone Project: CatchYou, a generative AI service that creates composite sketches from witness descriptions
- Leadership: Secretary General, Student Council

RED School – Alternative School
Geumsan, S.Korea

Student — *Mar. 2012 - Jan. 2018*

- Scholarship: Academic Excellence Scholarship (two consecutive years)
- Leadership: Student Council President

Experience

Artificial Intelligence and Data Analytics Lab, Myongji University
Seoul, S.Korea

Graduate Researcher — *Sep. 2024 - Aug. 2026*

- Developed scCap, a deep learning framework that predicts patient phenotypes from single-cell data without manual cell type annotation, using knowledge-augmented clustering (first-author paper in Briefings in Bioinformatics).
- Developed a multimodal fusion framework for COVID-19 severity prediction that combines clinical records, chest X-rays, and radiology reports and is extensible to new modalities (with KDCA).
- Developed deep learning models that classify tumor subtypes from single-cell data and applied them to drug target discovery (with AIgenDrug).
- Built an LLM-based AI agent environment (Biomni) to automate genomic data analysis for a CRISPR genome-editing analysis platform.
- Set up a scalable genome analysis workflow for apple whole-genome data using the Nextflow nf-core/sarek pipeline (with KNU).
- Analyzed spatial gene-expression data from breast cancer tissue to identify cell neighborhoods and the tumor microenvironment (with SNUH).
- Compared gene-expression patterns between liver cancer tissues with and without cirrhosis using spatial transcriptomics data.

Artificial Intelligence and Data Analytics Lab, Myongji University
Seoul, S.Korea

Undergraduate Research Intern — *Dec. 2022 - Aug. 2024*

- Developed ML models predicting in-hospital mortality of ARDS patients from ICU records (MIMIC-III), comparing missing-data strategies.
- Identified hypertension-progression risk groups in hyperlipidemia patients from KoGES cohort data by designing lab-based labels and building XGBoost, CatBoost, and TabNet models (with KDCA).

Bigleader
Seoul, S.Korea

AI Intern (Youth AI Internship) — *Jul. 2022 - Oct. 2022*

- Automated daily collection of foreign customs notices with Selenium and validated Sentence-BERT-based keyword extraction and HS code matching for a supply-chain risk monitoring system (with KOTRA).
- Received the KOTRA President's Merit Award and the TriBIG Project Scholarship.

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data
Briefings in Bioinformatics (BIB)

Published — *2026*

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids
Journal of Experimental & Clinical Cancer Research (JECCR)

Published — *2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq — *KIISE Korea Computer Congress (KCC)* 2026
Oral Presentation
Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques — *KIISE Korea Software Congress (KSC)* 2025
Poster Presentation (Best Poster Award)
Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data — *International Conference on Genome Informatics ISCB-Asia* 2025
Poster Presentation
J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning — *BIOINFO 2025, Korean Society for Bioinformatics (KSBI)* 2025
Poster Presentation (Excellent Poster Award)
J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning — *KIISE Korea Computer Congress (KCC)* 2025
Oral Presentation (Best Presentation Award)
J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies — *KIISE Korea Software Congress (KSC)* 2023
Poster Presentation
J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2026	**Research Honor Award**, Myongji University	*Seoul, S.Korea*
2025	**Best Poster Award**, KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award**, BIOINFO 2025, Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Best Presentation Award**, KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**President's Merit Award**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship**, Myongji University	*Seoul, S.Korea*
2018	**Scholarship III**, Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*

Projects

2024 – 2026	AI-based target discovery platform for CAR-Treg cell therapy in chronic inflammatory disease
2024 – 2026	Genetic and physiological mechanisms of apple blossom dormancy release under climate change
2024 – 2026	Spatial transcriptomics-based target discovery and validation for refractory solid tumors
2025 – 2026	Client-side web platform for detecting large CRISPR-Cas9-induced DNA deletions
2025 – 2026	GeoMx spatial transcriptomics of cirrhotic vs. non-cirrhotic HCC tissue
2025	COVID-19 severity prediction using multimodal clinical and imaging data
2023 – 2024	Hypertension-progression risk groups in hyperlipidemia patients (KoGES cohort)
2023	CatchYou: criminal composite sketch generation from witness descriptions
2022	Customs notice analysis and monitoring system for global supply-chain crises

Technical Skills

Programming	Python, R
Frameworks	PyTorch, TensorFlow, Scikit-learn
Tools	Linux, Git, Docker, Nextflow
Research	Machine Learning, Deep Learning, Multimodal Learning
Languages	Korean (Native), English (Fluent)
Certifications	Engineer Information Processing (2023), SQL Developer – SQLD (2023)

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University — *Seoul, S.Korea*

AI Team Member — *Oct. 2022 - Dec. 2023*

- Engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Data Youth Campus, MSIT & KDATA — *Seoul, S.Korea*

Participant (Dongguk University) — *Jul. 2022 - Aug. 2022*

- Completed a hands-on data talent program run by the Ministry of Science and ICT and KDATA, hosted at Dongguk University
- Conducted practical machine learning and data analysis projects

DeepsDeeps, Myongji University — *Seoul, S.Korea*

Study Group Member — *Mar. 2022 - Jul. 2022*

- Studied deep learning fundamentals and paper implementations with peers

FAA Football Analytics Academy, Bepro — *Seoul, S.Korea*

Data Integration Course Participant — *Sep. 2021 - Dec. 2021*

- Completed a sports data analytics course on integrating and analyzing football match data

Coding 101, Myongji University — *Seoul, S.Korea*

Member — *Mar. 2018 - Jan. 2020*

- Studied programming fundamentals and worked on coding projects in a student club

International

Camino de Santiago (French Way) — *Spain*

Pilgrim — *Sep. 2026 - Oct. 2026*

Medical Mission Trip to the Philippines (GMMA) — *Manila, Philippines*

Volunteer — *Jan. 2024*

- Participated in medical volunteer activities with international healthcare professionals

International Work Camp in Iceland — *Reykjavik, Iceland*

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and intercultural exchange with international volunteers

Mission Trip to Vietnam — *Hanoi, Vietnam*

Volunteer — *Aug. 2018*

- Participated in intercultural exchange and community volunteer activities